Provisional estimate: Estimate evidence

Prices and rates are as supplied with this export. They are not reviewed or approved, and no design review approval covers them.

Boards: MCC1

Currency: JOD

Description	Qty	Unit	Unit price	Total
Circuit-breaker 4P C63 -Q1	1	pcs	not priced	-
Circuit-breaker 3P C32 -Q2	1	pcs	not priced	-
Residual current circuit-breaker 4P 40A 30mA -F1	1	pcs	not priced	-
ABB T2S160 MF 12.5 -Q4	1	pcs	not priced	-
ABB A30 -Q5	1	pcs	not priced	-
ABB TA25DU14 -F2	1	pcs	not priced	-
Circuit-breaker 1P C16 -Q3	1	pcs	not priced	-
Cable 4G1.5 Cu PVC -W2	0	m	not priced	-
Cable 3G2.5 Cu PVC -W1	0	m	not priced	-
Wiring and testing, per circuit	2	pcs	10.00	20.00

Incomplete: these lines have no price and are not in the total:

circuit_breaker:4P:C63 (no price)

circuit_breaker:3P:C32 (no price)

residual_current_device:4P:40A:30mA (no price)

T2S160 MF 12.5 (no price)

A30 (no price)

TA25DU14 (no price)

circuit_breaker:1P:C16 (no price)

cable:4G1.5:Cu:PVC (no length)

cable:3G2.5:Cu:PVC (no length)

Materials	0.00
Labour and enclosure	20.00
Markup	0.00
VAT	0.00
Total (JOD)	**20.00**

Findings (0 blocking, 8 warning, 6 info)

[WARNING] MCC1 / c1: Pump: Ir 11.5 A, the typical Ir of a 5.5 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source: policy;

engineering confirmation required)

[WARNING] MCC1: Phase imbalance 55.6 % exceeds the company's 10 %: the loads cannot be spread more evenly as given. (phase_imbalance; source: company max_phase_imbalance_percent; engineering confirmation required)

[WARNING] MCC1: No fault level given: every breaker's breaking capacity is left to be confirmed. (no_fault_level; source: policy; engineering confirmation required)

[WARNING] MCC1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)

[WARNING] MCC1: 2 circuit(s) have no cable length given; their voltage drop is not checked. (voltage_drop_unchecked; source: policy; engineering confirmation required)

[WARNING] MCC1: No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze. (earth_fault_no_ze; source: policy;

engineering confirmation required)

[WARNING] MCC1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)

[WARNING] MCC1: Enclosure: none named, since 3 devices have no width (rail_widths_mm in the company settings). (enclosure_unknown_widths; source: policy; engineering confirmation required)

[INFO] MCC1: Group 1 breaker raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)

[INFO] MCC1 / c1: Pump: dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2. (starter_table; source: policy)

[INFO] MCC1: Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination. (discrimination_incomer_raised; source: policy)

[INFO] MCC1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)

[INFO] MCC1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip

(about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)

[INFO] MCC1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)